MATE 6551: Tarea 2

Due on November 29, 2025

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Problem 0

If $f : X \to Y$ is nullhomotopic and if $g : Y \to Z$ is continuous, then $g \circ f$ is nullhomotopic.

Proof:

Suppose that $f : X \to Y$ is nullhomotopic and $g : Y \to Z$ is continuous. Then there exists a homotopy $f \simeq k$ for some constant function $k : X \to Y$. Let $F : X \times I \to Y$ be such a homotopy. Then define $H : X \times I \to Z$ by $H(x,t) = g(F(x,t))$. Note that H is a composition of continuous functions and is, therefore, continuous. Moreover, $H(x,0) = g(F(x,0)) = g(f(x)) = (g \circ f)(x)$, and $H(x,1) = g(F(x,1)) = g(k(x)) = (g \circ k)(x)$. But k is constant, so $g \circ k$ must be constant as well.

$\therefore$ $g \circ f$ is nullhomotopic.

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Problem 1

*Prove that $X \mapsto CX$ defines a functor **Top** $\to$ **Top**.*

Proof:

The cone over a space X is $CX = \frac{X \times I}{X \times \{1\}}$. This defines the behavior of C on objects, we will define its behavior on morphisms.

Let $f : X \to Y$ be a continuous map, then define $Cf : CX \to CY$ by $Cf[(x,t)] = [(f(x),t)]$. To show that this map is well-defined, suppose that we have $[(x,t)] = [(x',t')]$ in CX, then $t = t'$ by construction of CX. If $t = t' = 1$, then:

$$Cf[(x,t)] = [(f(x),t)] = [(f(x),1)] = [(f(x'),1)] = [(f(x'),t')] = Cf[(x',t')] \tag{1}$$

If not, then $x = x'$, again by construction of Cf. Thus,

$$Cf[(x,t)] = [(f(x),t)] = [(f(x'),t')] = Cf[(x',t')] \tag{2}$$

Therefore, Cf is well-defined.

Let $\pi : Y \times I \to CY$ denote the quotient application, and $c : CX \to X \times I$ be a continuous choice function picking out a representative for every class. Then $Cf = \pi \circ (f \times 1_I) \circ c$, a composition of continuous functions. Therefore Cf is continuous.

We now show that C preserves compositions. Let $g : Y \to Z$ be a continuous map. Then:

$$\begin{aligned} C(g \circ f)[(x,t)] &= [((g \circ f)(x),t)] = [(g(f(x)),t)] \\ &= Cg[(f(x),t)] = Cg(Cf[(x,t)]) = (Cg \circ Cf)[(x,t)] \end{aligned} \tag{3}$$

Furthermore, it preserves identities:

$$C1_X[(x,t)] = [(1_X(x),t)] = [(x,t)] = 1_{CX}[(x,t)] \tag{4}$$

$\therefore$ C is a functor.

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Problem 2

If X and Y are path connected, then $X \times Y$ is path connected.

Proof:

Suppose that X and Y are path connected, and take $(x, y), (x', y') \in X \times Y$. Then $x, x' \in X$ and $y, y' \in Y$, but X and Y path connected $\Longrightarrow \exists f : I \to X, g : I \to Y$ paths from x to x' and from y to y' respectively. Now define $\varphi : I \to X \times Y$ to be $\varphi = f \times g$, that is, $\varphi(t) = (f(t), g(t))$. This map is clearly continuous, and satisfies:

$$\varphi(0) = (f(0), g(0)) = (x, y) \quad \text{and} \quad \varphi(1) = (f(1), g(1)) = (x', y') \tag{5}$$

Thus φ is a path from (x, y) to (x', y').

$\therefore$ $X \times Y$ is path connected.

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Problem 3

If $f : X \to Y$ is continuous and X is path connected, then $f(X)$ is path connected.

Proof:

Suppose that $f : X \to Y$ is continuous and that X is path connected. Take $f(x), f(x') \in f(X)$. Then $x, x' \in X$, but X path connected $\Longrightarrow \exists g : I \to X$ a path from x to x' in X. Defining $\varphi : I \to f(X)$ as the composite $\varphi = \tilde{f} \circ g$, where $\tilde{f} : X \to f(X)$ denotes the co-restriction of f to its image, yields a path from $f(x)$ to $f(x')$. Indeed, φ is clearly continuous, and:

$$\varphi(0) = \tilde{f}(g(0)) = \tilde{f}(x) = f(x), \quad \varphi(1) = \tilde{f}(g(1)) = \tilde{f}(x') = f(x') \tag{6}$$

$\therefore$ $f(X)$ is path connected.

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Problem 4

The path components of a space X are maximal path connected subspaces; moreover, every path connected subset of X is contained in a unique path component of X.

Proof:

We recall that, by definition, a path component of a space X is an equivalence class of points of X under the equivalence relation given by $x \sim y \Longleftrightarrow$ there exists a path from x to y.

Let X be a topological space and $[x] \subseteq X$ a path component. Suppose that $Y \subseteq X$ is a path connected subspace of X containing $[x]$, and take $y \in Y$. Since $[x] \subseteq Y$, we know that there must be a path from x to y, but that implies that $x \sim y$ and thus $y \in [x]$. Therefore, $Y = [x]$.

$\therefore$ $[x]$ is maximal with respect to path connected subspaces of X.

Let $Z \subseteq X$ be a path connected subspace of X, and take $z_0 \in Z$. Path connectedness implies that, for every $z \in Z$, there exists a path from z to z_0. That is,

$$z \sim z_0 \ \forall z \in Z \Longrightarrow z \in [z_0] \ \forall z \in Z \Longrightarrow Z \subseteq [z_0] \tag{7}$$

Therefore, Z is contained in a path component. For uniqueness, suppose that $[z_0], [z_1]$ are path components with $Z \cap [z_0] \neq \emptyset$ and $Z \cap [z_1] \neq \emptyset$. Then $\exists\ p_0 \in Z \cap [z_0]$, and $p_1 \in Z \cap [z_1]$. But Z is path connected, so there must be some path from p_0 to p_1. That is, $p_0 \sim p_1$, but we had $p_0 \in [z_0]$ and $p_1 \in [z_1]$. Then by transitivity, $z_0 \sim p_0 \sim p_1 \sim z_1 \Longrightarrow [z_0] = [z_1]$.

$\therefore$ Z is contained in a unique path component of X.

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